New Project: Generalised Spectral Ranking

Purpose

Generalise the EconomicsBusiness spectral ranking to cover all fields, producing source/institution rankings for world research output by oax topic/field.

Core abstraction: seed sets

A **set** defines the left-side nodes of a bipartite ranking. Two types, fully parallel:

type	left entities	left ← work join	output
`journal`	journals {S} in a field	`work.source_id` (one per work)	v_S, v_I
`institution`	institutions {I} in a field	`work.field_ids ∩ {T}` (multiple per work)	v_T, v_I

The bipartite spectral machinery (build_csr, power iteration, alpha/chi) is **identical** for both types. Only the edge-construction step differs.

These sets are constructed on a per-field basis. Your plan will include an approach to this field index that manages the data volume and processing time - an SQL scan of works/topics will be slow and ideally carried out for all fields simultaneously.

Scale

- ~2000 institutions
- ~2000 journals

Data sources

- OpenAlex citation data (existing pipeline)
- `works_topics.parquet` — (work_id, topic_id) — already constructed from OA Leiden topics

PROF

Topic set membership is resolved at runtime: `topic_ids WHERE field_id = X`.
No manual enumeration needed.

Config files

Use dataclasses to replace earlier approaches using csv to enhance extensiblity
It is unclear how to do this. It is part of your planning

Data persistence

Intermediate data structures are part of your planning. Persist expensive SQL.
Probably best to use tiered parquets (no DuckDB required at this scale):

```

```

```
data/
  edge_lists/   el_{set_id}_{window}.parquet      # heavy; one per
set×window
  units/        units_{set_id}_{window}.parquet
  rankings/     rankings_all.parquet              # small; set_id,
entity_type,
                                                  # entity_id, v, rank,
params
```

Rankings output is small (~1M rows total); keep as a single wide parquet for easy cross-set queries with pandas/polars.

What carries over from EconomicsBusiness

- `spectral_ranking/` — ranking engine (minor generalisation: `left_col` parameter)
- `prepare_data/` — OpenAlex ingestion pipeline
- `util/` — load_config, Paths, load_runs patterns

What has been jettisoned

- Field-specific corpus logic (field_eb, econ/business classification)
- `params.csv` schema
- All LaTeX / figure scripts
- vartau/fixtau distinction (may re-emerge but not a starting assumption)

Once edge construction is complete everything downstream is unchanged.